

TOPICS COVERED: *Function of a complex variable, Limit, Continuity, Differentiability, Analyticity, Analytic functions, Cauchy – Riemann equations (Cartesian and Polar form)*

4.1 PRELIMINARIES

A complex number is defined as an ordered pair (x, y) of real numbers, where $x \in \mathfrak{R}$, $y \in \mathfrak{R}$ and $\mathfrak{R}$ is the set of real numbers. If we write $z = (x, y)$, then $x = \text{Real part of } z \text{ (Re}(z))$ and $y = \text{Imaginary part of } z \text{ (Im}(z))$.

- A complex number is said to be purely real if its imaginary part is zero and purely imaginary if its real part is zero.
- $z_1 = (x_1, y_1)$, $z_2 = (x_2, y_2)$, $z_1 + z_2 = (x_1, y_1) + (x_2, y_2) = (x_1 + x_2, y_1 + y_2)$.

$$z_1 z_2 = (x_1, y_1)(x_2, y_2) = (x_1 x_2 - y_1 y_2, x_1 y_2 + y_1 x_2)$$

So, in general $(x, y) = (x, 0) + (0, y) = (x, 0) + (0, 1)(y, 0)$, where $(0, 1) \rightarrow i$

Thus, a complex number $z = (x, y) = x + iy$

$i^2 = (0, 1)(0, 1) = (-1, 0) = -1$ and i is regarded as the square root of real number -1 .

Complex numbers can be represented in the $x - y$ plane where x -axis is called the real axis and y -axis is called the imaginary axis. The $x - y$ plane in case of complex numbers is called the complex plane or Argand diagram. In the Argand diagram, the complex number z is represented by the point $P(x, y)$.

If (r, θ) are the polar coordinates of a point P , then the non-negative real number

$r = \sqrt{x^2 + y^2}$ is called the modulus of the complex number and denoted by $|z|$. In vector representation, the modulus represents the length of the vector $z = x + iy$. $\theta = \tan^{-1}\left(\frac{y}{x}\right)$ is

called the argument (or amplitude) of z and is denoted by $\arg(z)$ or $\text{amp}(z)$. It may be

noted that the argument of a complex number is not unique, since if θ be a value of the argument, so also is $2n\pi + \theta$, where n is any integer. Thus, the value of θ which satisfies the inequality $-\pi < \theta \leq \pi$ is called the principal value of the argument. Every non – zero complex number z can be expressed as

$$z = r(\cos \theta + i \sin \theta)$$

and the above form is called modulus – argument or polar or trigonometric form of z .

By Euler's formula, $\cos \theta + i \sin \theta = e^{i\theta}$, so above form can be written as

$$z = r(\cos \theta + i \sin \theta) = re^{i\theta},$$

and is called exponential form of z .

If $z = x + iy$ is any complex number, then the complex number $\bar{z} = x - iy$ is called the conjugate of the complex number z and is denoted by $\bar{z}$.

Properties of Complex Numbers

- $|z_1 - z_2| \rightarrow$ represents distance between two points z_1 and z_2 .
- $|z - z_0| = \rho$ represents a circle of radius ρ with centre at z_0
 $|z - z_0| < \rho$ represents interior or open region (open disk)
 $|z - z_0| \leq \rho$ represents closed region (closed disk)
 $|z - z_0| > \rho$ represents exterior region.
- Parametric representation of a circle: $z = z_0 + \rho e^{i\theta}$, $(0 \leq \theta \leq 2\pi)$ with centre at z_0 and radius ρ .
- The set of points which lie between two concentric circles $C_1: |z - z_0| = r_1$ and $C_2: |z - z_0| = r_2$ defines an open annulus i.e. the set of points which satisfies the

inequality $r_1 < |z - z_0| < r_2$. The set of points which satisfies the inequality $r_1 \leq |z - z_0| \leq r_2$ defines a closed annulus.

4.2 FUNCTION OF A COMPLEX VARIABLE: LIMIT, CONTINUITY AND DIFFERENTIABILITY

Function: If for each value of the complex variable $z(=x+iy)$ in a given set S of complex numbers, there exist one or more values of $w(=u+iv)$ by a given rule f , then w is said to be function of z , that is

$$w = f(z)$$

or
$$w = u(x, y) + iv(x, y) = f(z) = f(x + iy)$$

where u, v are real functions of x and y .

e.g. $f(z) = z^2$ for $z = x + iy$, we have

$$f(x + iy) = (x + iy)^2 = x^2 - y^2 + 2ixy = u(x, y) + iv(x, y).$$

To represent the function $w = f(z)$, we need a plane to represent each of the complex variable w and z . The sets of corresponding points $z(=x+iy)$ and $w(=u+iv)$ on the two planes as z -plane and w -plane. There exists a point (u, v) in the w -plane corresponding to each point (x, y) in the z -plane.

Single Valued Functions: If to each value of z , there corresponds only one value of w in a given set S , then it is called single - valued function e.g. $f(z) = z^2$.

Multiple Valued Functions: If to each value of z , there corresponds two or more values of w in a given set S , then it is called multiple - valued function. e.g. $f(z) = z^{1/2}$, $f(z) = z^{1/3}$. A multiple valued function can be considered as a collection of

single valued functions in which each member is called the branch of the function. In general, by function we always mean that it is single - valued.

Neighbourhood of a point: A neighbourhood (nbd.) of a point z_0 consists of all points lying inside, but not on a circle centered at z_0 and with a specific positive radius δ . Mathematically, a neighbourhood of point z_0 in the Argand plane is the set of all points z such that $|z - z_0| < \delta$, where δ is an arbitrary small positive number.

Deleted Neighbourhood: If from a neighbourhood (nbd.) of point z_0 , the point z_0 itself is deleted or excluded, we get a deleted neighbourhood. It is represented as, $0 < |z - z_0| < \delta$, that is, deleted nbd. includes all points z in a nbd. of z_0 excluding z_0 itself.

Limit: If $f(z)$ be defined and single valued in a nbd. of $z = z_0$, with the possible exception at $z = z_0$ (that is, in a deleted nbd. δ of z_0), then the number w_0 is the limit of $f(z)$ as z approaches z_0 and is written as,

$$\lim_{z \rightarrow z_0} f(z) = w_0$$

This means that the point $w = f(z)$ can be made arbitrarily close to w_0 if we choose the point z^* close enough to z_0 but distinct from it.

Mathematically,

$$\lim_{z \rightarrow z_0} f(z) = w_0,$$

means that for each positive number ε , $\exists$ a positive number δ such that,

$$|f(z) - w_0| < \varepsilon \text{ whenever } 0 < |z - z_0| < \delta,$$

Geometrically, the above definition means that, for each ε -neighbourhood, $|w - w_0| < \varepsilon$ of w_0 , there is a deleted δ neighbourhood $0 < |z - z_0| < \delta$ of z_0 such that every point z in it has an image w lying in the ε -neighbourhood.

Note: The limit must be independent of the manner in which z approaches z_0 and hence, it is always unique.

Continuity: Let $f(z)$ be defined and single valued in a nbd. of $z = z_0$ (that is, at z_0 also), then the function $f(z)$ is said to be continuous at $z = z_0$, if $\lim_{z \rightarrow z_0} f(z) = f(z_0)$.

This implies that the following three conditions should be satisfied for $f(z)$ to be continuous at $z = z_0$.

- i) $\lim_{z \rightarrow z_0} f(z) = w_0$ must exist.
- ii) $f(z_0)$ must exist, that is $f(z)$ is defined at z_0 .
- iii) $w_0 = f(z_0)$.

e.g. $f(z) = \begin{cases} z^2, & z \neq i \\ 0, & z = i \end{cases}$,

Mathematically, a function $f(z)$ is said to be continuous at a point z_0 , if for each positive number ε , $\exists$ a positive number δ such that,

$$|f(z) - f(z_0)| < \varepsilon \text{ whenever } |z - z_0| < \delta,$$

Note:

- i) A function $f(z)$ is said to be continuous in a region or domain D if it is continuous at each point $z \in D$.
- ii) Points in the z - plane where $f(z)$ fails to be continuous are called discontinuities of $f(z)$ and $f(z)$ is said to be discontinuous at these points. If $\lim_{z \rightarrow z_0} f(z)$ exists but it is not equal to $f(z_0)$, then z_0 is called a removable discontinuity (simple or jump discontinuity), since by redefining $f(z_0)$ to be the same as $\lim_{z \rightarrow z_0} f(z)$, the function can again become continuous.

- iii) If $f(z)$ is continuous in a region, then the real and imaginary parts of $f(z)$ are also continuous in the region, that is

$$w = f(z) = u + iv,$$

then if $f(z)$ is continuous, then so u and v .

- iv) The functions continuous in every finite region are i) all polynomials, ii) $\sin z, \cos z$, iii) e^z .

Differentiability: If $f(z)$ be a single valued function of a complex variable z defined in some region R of the z -plane. Then, $f(z)$ is said to be differentiable at a point z_0 of R iff

$\lim_{z \rightarrow z_0} \frac{f(z) - f(z_0)}{z - z_0}$ (or $\lim_{\Delta z \rightarrow 0} \frac{f(z_0 + \Delta z) - f(z_0)}{\Delta z}$) exists finitely independent of the path in

which $z \rightarrow z_0$ (or $\Delta z \rightarrow 0$). This limit if it exists finitely, is called the differential coefficient or derivative of f with respect to z at $z = z_0$, that is,

$$(4.1) \quad f'(z_0) = \lim_{z \rightarrow z_0} \frac{f(z) - f(z_0)}{z - z_0}$$

If $z = z_0 + \Delta z$, then as $z \rightarrow z_0$, $\Delta z \rightarrow 0$, then

$$(4.2) \quad f'(z_0) = \lim_{\Delta z \rightarrow 0} \frac{f(z_0 + \Delta z) - f(z_0)}{\Delta z}$$

Method to Check Differentiability: To show that $f(z)$ is non - differentiable, try different paths for limiting values in equation (4.1) or (4.2). Convenient paths are along real and imaginary axis i.e. we can take Δz either wholly real or wholly imaginary.

$$\frac{dw}{dz} = \lim_{\Delta z \rightarrow 0} \frac{\Delta w}{\Delta z}$$

For the differentiability of the function at the origin, the radius vector along the origin $y = mx$ is also a convenient path.

Exercise 4.1

4.1: Prove that $\lim_{z \rightarrow 0} \frac{\bar{z}}{z}$ does not exist.

4.2: For what value of z are each of the following functions are discontinuous

a) $f(z) = \frac{z}{z^2 + 1}$

b) $f(z) = \operatorname{cosec} z$

4.3: Show that the function $f(z) = |z|^2$ is continuous but not a differentiable function for any nonzero value of z .

Note: The example $f(z) = |z|^2$ shows that a function can be differentiable at a point but nowhere else in any nbd. of that point. This example also shows that $f(z)$ is continuous at each point in the plane but it is not differentiable at every point, so it shows that continuity does not imply the existence of differentiability. But, it is always true that differentiability implies the existence of continuity.

4.3 ANALYTIC FUNCTIONS AND CAUCHY - RIEMANN EQUATIONS

If $f(z)$ is single valued function and possess a unique derivative $f'(z)$ at all points of a region R , then $f(z)$ is said to be analytic function of z in that region R .

Analytic functions are also referred as "holomorphic" or "regular" or "monogenic" functions.

Analytic function at a point: If a function $f(z)$ is single - valued function defined in some region R , then $f(z)$ is said to be analytic at point z_0 of R if it is differentiable not only at z_0 but also in some nbd. of z_0 i.e. there exists a nbd. $|z - z_0| < \delta$ at all points of which $f'(z)$ exists.

Examples

- i) $f(z) = \frac{1}{z}$ is analytic at each non - zero point in the finite plane.
- ii) $f(z) = |z|^2$ is not analytic at any point since its derivative exists only at $z = 0$ and not throughout any nbd.(since smaller the circle we can take about $z = 0$, there should be non - zero points in it at which the function is non differentiable).

Entire Function: A function that is analytic at each point in the entire finite plane is referred as entire function.

e.g. polynomial $P(z) = a_0 + a_1z + a_2z^2 + \dots + a_nz^n, a_n \neq 0$ is an entire function since its derivative of polynomials exists everywhere.

Cauchy - Riemann equations in Cartesian Coordinates

For the function $f(z) = u(x, y) + iv(x, y)$,

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$$
$$\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

are Cauchy - Riemann (C R) equations in Cartesian form.

Necessary and Sufficient Condition for $f(z)$ to be analytic.

I) Necessary Condition for $f(z)$ to be Analytic

Theorem 4.1: *If the single - valued function $f(z) = u(x, y) + iv(x, y)$ is differentiable*

(analytic) at any point $z = x + iy$, then the partial derivatives $u_x \left(\frac{\partial u}{\partial x} \right), v_x \left(\frac{\partial v}{\partial x} \right), u_y \left(\frac{\partial u}{\partial y} \right),$

$v_y \left(\frac{\partial v}{\partial y} \right)$, should exist and satisfy the Cauchy - Riemann equations, that is,

$$u_x = v_y$$

and

$$u_y = -v_x$$

Proof: If

$$(4.3.1) \quad w = f(z) = u(x, y) + iv(x, y)$$

where $z = x + iy$.

Let the small increment in z be Δz , which results small increment Δx and Δy in x and y respectively, so

$$(4.3.2) \quad \Delta z = \Delta x + i\Delta y$$

Since the function is differentiable at any point z , therefore limit given by

$$(4.3.3) \quad \lim_{\Delta z \rightarrow 0} \frac{\Delta w}{\Delta z} = \lim_{\Delta z \rightarrow 0} \frac{f(z + \Delta z) - f(z)}{\Delta z}$$

must exist uniquely as $\Delta z \rightarrow 0$ along any path we choose.

Using relations (4.3.1) and (4.3.2), the relation (4.3.3) can be written as:

$$(4.3.4) \quad \lim_{\Delta z \rightarrow 0} \frac{\Delta w}{\Delta z} = \lim_{(\Delta x, \Delta y) \rightarrow 0} \frac{\{u(x + \Delta x, y + \Delta y) + iv(x + \Delta x, y + \Delta y)\} - \{u(x, y) + iv(x, y)\}}{\Delta x + i\Delta y}$$

Considering the RHS of above equation

$$(4.3.5) \quad \lim_{(\Delta x, \Delta y) \rightarrow 0} \frac{\{u(x + \Delta x, y + \Delta y) - u(x, y)\}}{\Delta x + i\Delta y} + i \frac{\{v(x + \Delta x, y + \Delta y) - v(x, y)\}}{\Delta x + i\Delta y}$$

II) Sufficient Condition for $f(z)$ to be analytic.

Theorem 4.2: *The single valued continuous function $f(z) = u(x, y) + iv(x, y)$ is analytic in a region R , if the four partial derivatives $u_x \left(\frac{\partial u}{\partial x} \right)$, $v_x \left(\frac{\partial v}{\partial x} \right)$, $u_y \left(\frac{\partial u}{\partial y} \right)$, $v_y \left(\frac{\partial v}{\partial y} \right)$, exist, continuous and satisfy C - R equations at each point of R .*

Proof: Taking $w = f(z) = u(x, y) + iv(x, y)$ and Δz be increment in z with their corresponding increments Δx and Δy in x and y respectively i.e. $\Delta z = \Delta x + i\Delta y$. Thus,

$$(4.3.8) \quad \Delta w = \Delta u + i\Delta v$$

where $\Delta u = u(x + \Delta x, y + \Delta y) - u(x, y)$ and $\Delta v = v(x + \Delta x, y + \Delta y) - v(x, y)$.

In view of the continuity of first four partial derivatives of u and v at the point (x, y) , we can expand Δu and Δv by Taylor's theorem for functions of two variables up to first order terms as,

$$\Delta u = u(x + \Delta x, y + \Delta y) - u(x, y) = u(x, y) + \left(\Delta x \frac{\partial u}{\partial x} + \Delta y \frac{\partial u}{\partial y} \right) - u(x, y)$$

$$(4.3.9) \quad \Delta u = \Delta x \frac{\partial u}{\partial x} + \Delta y \frac{\partial u}{\partial y}$$

Similarly, we get

$$(4.3.10) \quad \Delta v = \Delta x \frac{\partial v}{\partial x} + \Delta y \frac{\partial v}{\partial y}$$

Now, with equations (4.3.9) and (4.3.10), equation (4.3.8) becomes

$$(4.3.11) \quad \Delta w = \{u_x(x, y)\Delta x + u_y(x, y)\Delta y\} + i\{v_x(x, y)\Delta x + v_y(x, y)\Delta y\}$$

Taken into consideration that the C - R equations are satisfied, we can use $u_y = -v_x$,

$v_y = u_x$ in equation (1.3.11) and we get,

$$\frac{\Delta w}{\Delta z} = \{u_x(x, y)\Delta x - v_x(x, y)\Delta y\} + i\{v_x(x, y)\Delta x + u_x(x, y)\Delta y\}$$

or

$$(4.3.12) \quad \frac{\Delta w}{\Delta z} = \frac{\{u_x(x, y) + iv_x(x, y)\}\Delta x + i\{u_x(x, y) + iv_x(x, y)\}\Delta y}{\Delta z} \\ = \frac{\{u_x(x, y) + iv_x(x, y)\}(\Delta x + i\Delta y)}{\Delta z}$$

so we get equation (4.3.12) as

$$\lim_{(\Delta x, \Delta y) \rightarrow (0, 0)} \frac{\Delta w}{\Delta z} = u_x + iv_x$$

Thus, we get (4.3.12) as,

$$f'(z) = \lim_{\Delta z \rightarrow 0} \frac{\Delta w}{\Delta z} = u_x + iv_x. \quad ###$$

Note:

$$i) \quad f'(z) = u_x + iv_x, \text{ so } |f'(z)| = \sqrt{u_x^2 + v_x^2}$$

$$\text{Taking } |f'(z)|^2 = u_x^2 + v_x^2 = u_x u_x + v_x v_x = u_x v_y - v_x u_y$$

So. we get $|f'(z)|^2 = \frac{\partial(u, v)}{\partial(x, y)}$ (represents the Jacobian of u and v with respect to independent variables x and y).

$$ii) \quad \text{If } w = u + iv, \text{ where } z = x + iy, \text{ then } f'(z) = \frac{dw}{dz} = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} = \frac{\partial}{\partial x} (u + iv) = \frac{\partial w}{\partial x}$$

iii) The real and imaginary parts of an analytic function are called conjugate functions.

Polar form of Cauchy - Riemann Equations

$$\text{The polar form of Cauchy Riemann equations are} \quad \frac{\partial u}{\partial r} = \frac{1}{r} \frac{\partial v}{\partial \theta} \\ \frac{\partial u}{\partial \theta} = -r \frac{\partial v}{\partial r}$$

Proof: As $x = r \cos \theta$ and $y = r \sin \theta$, we can write $z = r(\cos \theta + i \sin \theta) = re^{i\theta}$

If $w = f(z) = u + iv = f(re^{i\theta})$, so

$$(4.3.13) \quad u(r, \theta) + iv(r, \theta) = f(re^{i\theta})$$

Differentiating (4.3.13) partially w.r.t. r , we get

$$(4.3.14) \quad \frac{\partial u}{\partial r} + i \frac{\partial v}{\partial r} = f'(re^{i\theta})e^{i\theta}$$

Differentiating (4.3.13) partially w.r.t. θ , we get

$$(4.3.15) \quad \frac{\partial u}{\partial \theta} + i \frac{\partial v}{\partial \theta} = f'(re^{i\theta})rie^{i\theta}$$

Substituting the value of $f'(re^{i\theta})e^{i\theta}$ from (4.3.15) in equation (1.3.14), we get

$$\frac{\partial u}{\partial r} + i \frac{\partial v}{\partial r} = \frac{1}{r} \frac{\partial v}{\partial \theta} - i \frac{1}{r} \frac{\partial u}{\partial \theta}$$

or

$$\frac{\partial u}{\partial r} = \frac{1}{r} \frac{\partial v}{\partial \theta},$$

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$$\frac{\partial u}{\partial \theta} = -r \frac{\partial v}{\partial r}$$

Exercise 4.2

4.4: Test the analyticity of the function $w = \sin z$ and hence derive that $\frac{d}{dz}(\sin z) = \cos z$.

4.5: Find the values of a and b such that the function

$$f(z) = x^2 + ay^2 - 2xy + i(bx^2 - y^2 + 2xy) \text{ is analytic.}$$

Ans.: $a = -1, b = 1$.

4.6: Show that the function $f(z) = z|z|$ is non - analytic anywhere.

4.7: Show that the function defined by $f(z) = \sqrt{|xy|}$ satisfies C - R equations at the origin but it is not analytic at that point.

4.8: If $f(z) = \begin{cases} \frac{x^3 y(y - ix)}{x^6 + y^2}, & z \neq 0 \\ 0, & z = 0 \end{cases}$, prove that $\frac{f(z) - f(0)}{z} \rightarrow 0$ as $z \rightarrow 0$ along any radius

vector but not as $z \rightarrow 0$ in any manner and also that $f(z)$ is not analytic at $z = 0$.

4.9: Using Cauchy Riemann equations in polar form, deduce that:

$$\frac{\partial^2 u}{\partial r^2} + \frac{1}{r} \frac{\partial u}{\partial r} + \frac{1}{r^2} \frac{\partial^2 u}{\partial \theta^2} = 0$$